Raspberry Pi 3 RFID Kiosk Setup

This document summarizes the end-to-end process of configuring a Raspberry Pi 3 from scratch to automatically boot into a fullscreen Python RFID kiosk application.

1. Flash Raspberry Pi OS and Create User

Use the Raspberry Pi Imager to install the operating system onto an SD card:

- Select **Raspberry Pi 3** as the device.
- Select **Raspberry Pi OS (Legacy, 32-bit) with Desktop**. The legacy version uses X11, which is required for this setup.
- In OS Customization (gear icon), set the username exactly to `rentgen`.
- Configure Wi-Fi settings, save, and flash the SD card.

2. Configure Auto-Login

Ensure the Pi boots directly to the desktop without requiring a password prompt:

1. Open a terminal on the Raspberry Pi.
2. Run the configuration tool: `sudo raspi-config`
3. Navigate to **System Options** > **Boot** / **Auto Login**.
4. Select **Desktop Autologin**.
5. Exit the tool and reboot the Raspberry Pi.

3. Install Python Dependencies

Install the required system packages for Tkinter, Pillow (image manipulation), and Pygame (audio playback):

```
sudo apt update
sudo apt install python3-tk python3-pil python3-pil.imagetk python3-pygame
-y
```

4. Organize Project Files

Ensure all necessary files are located in the user's home directory (`/home/rentgen/`):

- `rentgen.py` (Ensure the script is named exactly this to match the service)
- `background.png`
- `scan_result.png`
- `beep.mp3`

Make the Python script executable:

```
chmod +x /home/rentgen/rentgen.py
```

5. Install and Enable the Service

Configure a systemd service to run the script automatically on boot.

1. Create the service file using nano:
   `sudo nano /etc/systemd/system/rfid_display.service`
2. Paste your provided unit configuration (ensuring `[Unit]` is properly bracketed on the first line).
3. Reload the systemd daemon:

```
sudo systemctl daemon-reload
```

Enable the service to start on boot:

```
sudo systemctl enable rfid_display.service
```

Start it immediately to test:

```
sudo systemctl start rfid_display.service
```

Official Resources & Links

- **Raspberry Pi Imager:** raspberrypi.com/software
- **Raspberry Pi Documentation:** raspberrypi.com/documentation
- **Pygame:** pygame.org
- **Pillow (PIL):** python-pillow.org
- **Python Tkinter:** docs.python.org/3/library/tkinter.html